

```

!pip install streamlit
import streamlit as st
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

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Requirement already satisfied: streamlit in /usr/local/lib/python3.12/dist-packages (1.54.0)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=4.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (5.5.0)
Requirement already satisfied: blinker<2,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (1.9.0)
Requirement already satisfied: cachetools<7,>=5.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.2.6)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.12/dist-packages (from streamlit) (8.3.1)
Requirement already satisfied: gitpython!=3.1.19,<4,>=3.0.7 in /usr/local/lib/python3.12/dist-packages (from streamlit) (3.1.46)
Requirement already satisfied: numpy<3,>=1.23 in /usr/local/lib/python3.12/dist-packages (from streamlit) (2.0.2)
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Requirement already satisfied: tornado!=6.5.0,<7,>=6.0.3 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.5.1)
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Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.12/dist-packages (from streamlit) (6.0.0)
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Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (4.17.0)
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.12/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (1.14.2)
Requirement already satisfied: gitdb<5,>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from gitpython!=3.1.19,<4,>=3.0.7->streamlit) (4.0.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<3,>=1.4.0->streamlit) (2.8.2)
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Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (3.0.2)
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Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (0.36.0)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=4.0->streamlit) (0.25.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<3,>=1.4.0->streamlit) (1.17.0)

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# =====
# KONFIGURASI HALAMAN
# =====
st.set_page_config(page_title="Dashboard Kepuasan Pegawai", layout="wide")
st.title("📊 Dashboard Mini Kepuasan Layanan Kepegawaian")
st.markdown("Analisis berbasis data untuk mendukung rekomendasi kebijakan")

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2026-03-01 13:28:42.160 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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DeltaGenerator()

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# =====
# LOAD DATA
# =====
df = pd.read_excel("data_survei_simulasi_10_responden.xlsx", engine="openpyxl")
# Ambil indikator V1-V5 dan pastikan numerik
indikator = df.iloc[:, 1:6].apply(pd.to_numeric, errors="coerce")

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# =====
# KPI KEPUASAN (IKM)
# =====
mean_scores = indikator.mean()
ikm = (mean_scores.mean() / 5) * 100

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def kategori_ikm(x):
    if x >= 81: return "Sangat Baik"
    elif x >= 66: return "Baik"
    elif x >= 51: return "Cukup"
    else: return "Kurang"

col1, col2, col3 = st.columns(3)
col1.metric("📊 Indeks Kepuasan (IKM)", f"{ikm:.2f}%")
col2.metric("🏷️ Kategori", kategori_ikm(ikm))
col3.metric("👤 Responden", len(df))

st.divider()
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DeltaGenerator()
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# =====
# 3 ANALISIS GAP
# =====
st.header("3 Analisis GAP (Expectation vs Performance)")

gap_scores = 5 - mean_scores
prioritas_gap = gap_scores.idxmax()

fig_gap, ax_gap = plt.subplots(figsize=(6,4))
ax_gap.bar(gap_scores.index, gap_scores.values, color=plt.cm.Set2(range(len(gap_scores))))
ax_gap.set_ylabel("Nilai GAP")
ax_gap.set_title("GAP Kepuasan per Indikator")
ax_gap.grid(axis="y", linestyle="--", alpha=0.6)

for i, v in enumerate(gap_scores.values):
    ax_gap.text(i, v + 0.03, f"{v:.2f}", ha="center", fontweight="bold")

st.pyplot(fig_gap)
st.info(f"🔴 Prioritas perbaikan tercepat: **{prioritas_gap}**")

st.divider()
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DeltaGenerator()
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# =====
# 4 ANALISIS KORELASI
# =====
st.header("4 Korelasi Antar Indikator")

corr = indikator.corr()

fig_corr, ax_corr = plt.subplots(figsize=(6,5))
im = ax_corr.imshow(corr, cmap="coolwarm", vmin=-1, vmax=1)
plt.colorbar(im, ax=ax_corr)
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ax_corr.set_xticks(range(len(corr.columns)))
ax_corr.set_yticks(range(len(corr.columns)))
ax_corr.set_xticklabels(corr.columns, rotation=45, ha="right")
ax_corr.set_yticklabels(corr.columns)

for i in range(len(corr)):
    for j in range(len(corr)):
        ax_corr.text(j, i, f"{corr.iloc[i, j]:.2f}", ha="center", va="center")

ax_corr.set_title("Heatmap Korelasi Pearson")
st.pyplot(fig_corr)

corr_v5 = corr.iloc[:-1, -1].sort_values(ascending=False)
st.subheader("📊 Ranking Faktor Berpengaruh")
st.dataframe(corr_v5.to_frame("Koefisien Korelasi"))

st.divider()

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2026-03-01 13:28:42.833 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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DeltaGenerator()

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# =====
# 📊 ANALISIS REGRESI
# =====
st.header("📊 Analisis Regresi Linear Berganda")

X = sm.add_constant(indikator.iloc[:, 0:4])
y = indikator.iloc[:, 4]

model = sm.OLS(y, X, missing="drop").fit()

coef = model.params[1:]
r2 = model.rsquared

fig_reg, ax_reg = plt.subplots(figsize=(6,4))
ax_reg.bar(coef.index, coef.values, color="#3498db")
ax_reg.axhline(0, linestyle="--", color="black")
ax_reg.set_title("Koefisien Regresi")

st.pyplot(fig_reg)
st.info(f"📊 Nilai R²: **{r2:.2f}**")
st.success(f"👉 Faktor dominan: **{coef.abs().idxmax()}**")

st.divider()

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2026-03-01 13:28:43.896 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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DeltaGenerator()

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```
# 6 SEGMENTASI KEPUASAN (FIX PALING AMAN)
# =====
st.header("🔍 Segmentasi Kepuasan Pegawai")

scaler = StandardScaler()
X_scaled = scaler.fit_transform(indikator.fillna(indikator.mean(

kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
cluster_label = kmeans.fit_predict(X_scaled)

# 🔥 GROUPBY PALING AMAN (HANYA DATA NUMERIK)
indikator_cluster = indikator.copy()
indikator_cluster["Cluster"] = cluster_label

cluster_mean = indikator_cluster.groupby("Cluster").mean()
cluster_mean = cluster_mean.sort_values(by=indikator.columns[-1])

cluster_mean["Segment"] = ["Sangat Puas", "Cukup Puas", "Tidak P
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2026-03-01 13:28:44.517 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-03-01 13:28:44.521 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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# Radar Chart
labels = indikator.columns.tolist()
angles = np.linspace(0, 2*np.pi, len(labels), endpoint=False).tolist()
angles += angles[:1]

fig_rad = plt.figure(figsize=(6,6))
ax_rad = plt.subplot(polar=True)

colors = ["#2ecc71", "#f1c40f", "#e74c3c"]

for i, row in cluster_mean.iterrows():
    values = row[labels].tolist() + [row[labels].tolist()[0]]
    ax_rad.plot(angles, values, label=row["Segment"], color=colors[i])
    ax_rad.fill(angles, values, alpha=0.25)

ax_rad.set_thetagrids(np.degrees(angles[:-1]), labels)
ax_rad.set_ylim(0, 5)
ax_rad.set_title("Radar Segmentasi Kepuasan")
ax_rad.legend(loc="upper right")

st.pyplot(fig_rad)

st.success("🎉 Segmentasi berhasil - siap untuk rekomendasi kebijakan")
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2026-03-01 13:28:44.633 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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DeltaGenerator()
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